

Yunqi Xu

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Education

Johns Hopkins University

B.S. IN COMPUTER SCIENCE, MINOR IN ROBOTICS

Maryland, MD, USA

Aug.2026 - May.2028

Research Interests

Core Research Interests: My research lies at the intersection of **biomimetic robotics**, **autonomous systems**, and **precision agriculture**. I am particularly focused on developing **bio-inspired solutions** to tackle the challenges of dynamic **agricultural environments**. My work spans from mechanical design to coordination.

Core Research Question: How can we integrate **biomimetic end-effectors** and **soft grasping with biomimetic sensing and active vision** to achieve delicate crop manipulation, and further scale these capabilities through biomimetic swarm intelligence and collaborative control for **large-scale, reliable agricultural automation**?

Honors and Awards (Selected)

- 2026 Spring 2026 Dean's list,
- 2025 Fall 2025 Dean's list,
- 2025 Spring 2025 Dean's list,
- 2024 Fall 2024 Dean's list,
- 2024 Utility Model Patent Certificate,

Minneapolis, MN, USA
Minneapolis, MN, USA
Minneapolis, MN, USA
Minneapolis, MN, USA
China

Project Portfolio

Ergonomic Infant Carrier Structural Design

LEAD DESIGNER AND INVENTOR

- Patent No. ZL202420025612.5, [\[link\]](#)
- Designed a novel force-distribution structure to reduce spinal pressure for users by **15%** based on biomechanical modeling.
- Utilized **MATLAB** to simulate load-bearing points and optimized the mechanical linkages of the carrier.
- Successfully filed and obtained a **Chinese Utility Model Patent** for the innovative structural improvement.

China
2024

Skills

Programming Python, C++, $\LaTeX$, JAVA, MATLAB, R, Mathematica

Outreach Experience

Cedar Riverside Adult Education Collaborative Volunteering

TUTOR

- Taught English to level 1 adult learners

Minneapolis, MN, USA
May.2025 - June.2025

ABC Kids Expo, [\[link\]](#)

SALE, TRANSLATOR(ENGLISH, CANTONESE, MANDARIN)

- Engaged with well-known stroller brands, such as Familidoo, and shared ideas about parenting solutions suitable for families with multiple children
- discussed onboarding the products to Amazon, Target, and Walmart

Las Vegas, NV, USA
May.21-23.2025